

monotropism-autism-prevalence-global

Autism and Monotropism: How Common Are They, Really?

A cited research brief — fact-checking the regional autism figures and the "what percentage of people are monotropic" question.

Informational, not medical advice. *These figures describe how many people are counted as autistic or score highly on a monotropism scale. Counting depends heavily on who is looking, how, and where. Nothing here is diagnostic.*

Executive summary

- **Autism is not 1% everywhere.** The cleanest global figure is the **Global Burden of Disease (GBD) 2021** estimate, now adopted by the WHO: about **1 in 127 people (~0.8%)** were on the autism spectrum in 2021 [1][2]. For *children*, pooled reviews still land near the long-standing **~1 in 100** [3]. The United States — the most intensively screened country — reports **1 in 31 (3.2%)** among 8-year-olds (CDC, 2022 data) [4].
- **Regional differences are mostly a measurement artefact, not biology.** The same GBD study found autism-related burden *lowest* in South-East/East Asia and Oceania and *highest* in the high-income super-region [1]. That pattern tracks diagnostic infrastructure, awareness and screening — not real differences in how often autism occurs.
- **The frequently-quoted regional table is substantially wrong for the Middle East and over-simplified for Asia.** A 2025 meta-analysis puts the Middle East & North Africa (MENA) pooled prevalence at only **~0.14%** (range 0.01% Oman → 2.51% Saudi Arabia) [6], and an Asian

meta-analysis reports East Asia ~**0.51%** and mainland China ~0.7% [5] [7] — far below the "0.9–1.5%" often claimed for the Arab world.

- **Monotropism has no agreed "prevalence," and the "15–20% of people are monotropic" figure is a category error.** Monotropism is a *dimensional* cognitive style (a continuous trait), not a diagnosis. The only large empirical study — the **Monotropism Questionnaire (MQ)**, Garau et al. 2023 — found autistic participants averaged **4.15/5** and non-autistic participants **3.19/5**, with substantial overlap [10]. It does **not** say "X% of people are monotropic."
- **The ">90% of autistic people are monotropic" claim is a theoretical statement about how well the theory captures autistic experience, not a measured percentage.** It is defensible as *subjective validity*, misleading as *epidemiology*.

1. How many people are autistic? (and why two headline numbers exist)

Two figures circulate, and they answer slightly different questions:

- **~1 in 100 children.** This is the figure pooled across dozens of studies in systematic reviews (e.g. Zeidan et al., 2022) and is the WHO's long-standing *children-focused* summary [3]. It reflects studies that mostly screen paediatric populations.
- **1 in 127 people (~0.79%), all ages.** This is the **GBD 2021** modelled estimate — 61.8 million people in 2021 — which the WHO has now adopted verbatim ("In 2021 about 1 in 127 persons had autism") [1][2]. It is lower per-capita because it spans the whole lifespan, including older generations who were never screened.

The gap between "1 in 100" and "1 in 127" is not a contradiction: the first is children, the second is everyone. Both are defensible if the population is stated.

The GBD also gives a clear sex breakdown — about **1.06% of males** versus **0.51% of females** (1,064.7 vs 508.1 per 100,000) — reflecting the well-documented under-identification of autistic girls and women, not a two-fold biological difference [1].

2. Why regional numbers differ (the differences mostly measure *detection*, not biology)

This is the single most important point for reading any regional table. The GBD 2021 authors found autism-related health burden ranged from **126.5 disability-adjusted life-years (DALYs) per 100,000** in the South-East Asia / East Asia / Oceania super-region to **204.1 per 100,000** in the high-income super-region [1]. In plain terms: the wealthier, more-screened parts of the world *find* far more autism than the rest.

The drivers of the gap are almost entirely methodological and social:

- **Diagnostic infrastructure** — active surveillance networks (e.g. the US CDC's ADDM) systematically case-find in schools and clinics; many countries have no such system and rely on passive clinical records.
- **Criteria and awareness** — the shift from narrow "autistic disorder" (DSM-IV) to the broader autism spectrum (DSM-5, 2013), plus growing public and professional awareness, has steadily increased *identified* prevalence in countries that re-screen.
- **Stigma, language and service access** — in many regions autism is under-reported because families face stigma, services are absent, or screening tools are not validated in the local language and culture.

So when a table shows "0.1% in region X and 3% in region Y," read it as "*region Y looks harder and has more capacity to diagnose,*" not "*autism is thirty times rarer in region X.*" The GBD authors stress exactly this: better global data coverage is needed before geographic variation can be interpreted confidently [1].

3. A corrected regional table

The figures that prompted this brief are reproduced against the best available evidence. "Best estimate" columns cite primary systematic reviews or the GBD model — not advocacy round-ups.

Region	Figure often quoted	What the evidence supports	Source
World — all ages	"1%"	1 in 127 (~0.79%) in 2021 (GBD; now WHO's figure)	[1][2]

Region	Figure often quoted	What the evidence supports	Source
World — children	"1 in 100"	~1 in 100 children (pooled reviews)	[3]
North America	"1–3.2%"	3.2% (1 in 31) US 8-year-olds (2022, CDC ADDM); ~1 in 45 adults	[4]
Europe	"0.7–1%"	Registry-based studies cluster around ~ 1% (e.g. Denmark, Finland, Iceland); population-based estimates somewhat higher	[3]
Middle East & North Africa	"0.9–1.5%"	Pooled ~ 0.14% (95% CI 0.02–0.36%); single-country range 0.01% (Oman) → 2.51% (Saudi Arabia)	[6]
East / South-East Asia	"0.4–1.4%"	Asian meta-analysis: East Asia ~ 0.51% , South Asia ~0.31%, West Asia ~0.35%; mainland China ~ 0.7% (1 in 143) ; highly-screened pockets (Singapore, Japan, S. Korea) report higher	[5][7]

Two takeaways from the table:

1. The **North American** and **European** ranges are broadly right — these are the most-screened regions, so their high numbers are expected.
2. The **MENA** figure is the big error. A pooled prevalence near **0.14%** almost certainly reflects *under-diagnosis and weak surveillance*, not a genuinely rarer autism. The enormous within-region spread (Oman 0.01% to Saudi Arabia 2.51%) is itself the tell: real biology does not vary that much between neighbouring countries; surveillance does [6].

4. Is autism actually "rising"?

Headline prevalence has climbed steeply — in the US from roughly 1 in 150 (2000) to 1 in 31 (2022) [4]. The dominant explanation among epidemiologists is **better and broader identification**, not necessarily more autism:

- broadened criteria (DSM-IV → DSM-5),
- active case-finding (ADDM-style surveillance),
- earlier and more equitable screening across sex, race and ethnicity,

- reduced stigma leading to more assessment.

The GBD authors note prevalence has been comparatively *stable* where data quality is controlled, and that apparent "increases" largely track detection capacity [1]. The honest summary: **we are increasingly counting autism that was always there.**

5. How many people are "monotropic"? (the question that has no single answer)

Monotropism (Murray, Lawson & Lesser, 2005) is an *attention-based* account of autism: a monotropic mind concentrates its resources into a few intensely-aroused "attention tunnels," rather than spreading attention across many channels (polytropism). Crucially, it is framed as a **difference in attention allocation, not a deficit** — and as a **spectrum**, not a yes/no category.

Because of that, there is **no agreed prevalence of "monotropism"**, for three structural reasons:

1. **It is not a diagnosis.** Monotropism appears nowhere in the DSM-5 or ICD-11. There is no case register and no diagnostic threshold.
2. **It is dimensional, not categorical.** Everyone can enter a focused "flow" state; the autistic difference is that deep, narrow focus is the *default operating mode* rather than an occasional state. A line between "monotropic" and "polytropic" would be arbitrary.
3. **The measure reports means, not cases.** The validated instrument — the **Monotropism Questionnaire (MQ)**, Garau et al. 2023 — yields a continuous score from 1 to 5, and the researchers compare *group averages*, not "how many people exceed a cutoff" [10].

The actual numbers we do have (MQ validation study)

In the MQ development study (1,110 participants: 756 autistic, 354 non-autistic) [10]:

- **Autistic participants: mean 4.15 / 5 (SD 0.347)**
- **Non-autistic participants: mean 3.19 / 5 (SD 0.578)**
- **ADHD status** independently predicted higher monotropism scores (relevant because autism and ADHD co-occur in roughly 30–80% of cases depending on direction).

- An exploratory factor analysis produced an **8-factor structure** (special interests; rumination/anxiety; need for routines; environment & the attention tunnel; losing track of surroundings; decision-making; anxiety-reducing effect of interests; managing social interactions).

The overlap between the two distributions is substantial — many non-autistic people score above some autistic people's scores, and vice versa. That overlap is *exactly* why a "% of the population is monotropic" headline is not meaningful without an arbitrary (and contested) cutoff.

6. Where the ">90%" and "15–20%" figures come from — and why they are not prevalence data

These two numbers, which circulate widely online, deserve a direct fact-check.

">90% of autistic people are monotropic." This is best read as a claim about **subjective validity** — that is, *when autistic people encounter the monotropism framing, an overwhelming majority recognise it as an accurate description of their experience*. That is a real and meaningful phenomenon (the MQ was built *with* an autistic-led community group precisely because the description resonated [10]). But it is **not an epidemiological prevalence**; no study has computed "the proportion of autistic people who are monotropic" against a validated threshold, because there is no agreed threshold. Stated as *"monotropism has very high subjective validity for autistic people,"* it is defensible; stated as *"over 90% of autistic people are monotropic,"* it overstates what has been measured.

"15–20% of the general population is monotropic." This figure is almost certainly a **conflation with "neurodivergence."** Community estimates put *some* form of neurodivergence (ADHD, dyslexia, dyspraxia, Tourette's, etc., with heavy overlap) in the order of 15–20% of the population. For context, even the better-measured components are much smaller in isolation: adult ADHD affects about **2.58%** (persistent) to **6.76%** (symptomatic) of adults globally [8], and childhood ADHD is commonly estimated around 5–8%. Dyslexia is often quoted near ~10% depending on definition. **None of these is a monotropism figure.** Equating "neurodivergent" with "monotropic" is a category error: monotropism is one *dimension* of cognitive style that is elevated in — but

not identical to — several neurodivergent profiles. There is currently **no evidence-based number** for the prevalence of monotropism in the general population.

Bottom line on monotropism: *treat it as a measurable tendency on a spectrum, strongly (but not exclusively) associated with autism and ADHD, with high lived-experience validity for autistic people. The honest answer to "what percentage of people are monotropic?" is: that is not yet a well-defined question, and any precise percentage you see is invented.*

7. Practical takeaways

- **Quote the figure that matches the population.** "1 in 127 people" (all ages, GBD/WHO) and "~1 in 100 children" are both fine — name which one you mean [1][2][3].
 - **Treat regional gaps as detection gaps.** Low reported prevalence in MENA, parts of Asia, and among girls/women, older adults, and racialised groups reflects under-identification, not absence [1][6].
 - **Don't present monotropism as having a prevalence.** Report MQ means and the dimensional framing; avoid ">90%" and "15–20%" as prevalence statements [10].
 - **Keep the framing neuroaffirming.** Higher autism *identification* in well-resourced countries is a success of awareness and access; monotropic vs polytropic is a *difference in attention*, not a deficit — consistent with how the originators and the autistic community frame the theory.
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Sources

[1] Global Burden of Disease Study 2021 Autism Spectrum Collaborators (2024). *The global epidemiology and health burden of the autism spectrum: findings from the Global Burden of Disease Study 2021*. The Lancet Psychiatry (19 Dec 2024). doi:10.1016/S2215-0366(24)00363-8 — 61.8M people (1 in 127) in 2021; age-standardised prevalence 788.3/100,000 (males 1,064.7; females 508.1); super-region DALYs 126.5 (SE/E Asia & Oceania) to 204.1 (high-income) per 100,000.

[https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366\(24\)00363-8/fulltext](https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366(24)00363-8/fulltext) · IHME summary:

<https://www.healthdata.org/research-analysis/library/global-epidemiology-and-health-burden-autism-spectrum-findings-global>

[2] World Health Organization (2024). *Autism fact sheet (key facts)*. — "In 2021 about 1 in 127 persons had autism." <https://www.who.int/news-room/fact-sheets/detail/autism-spectrum-disorders>

[3] Zeidan, J. et al. (2022). *Global prevalence of autism: A systematic review update*. Autism Research (Wiley). doi:10.1002/aur.2696 — ~1 in 100 children; European registry data (France, Denmark, Finland, Iceland). <https://onlinelibrary.wiley.com/doi/full/10.1002/aur.2696>

[4] CDC, Autism and Developmental Disabilities Monitoring (ADDM) Network (2025). *Prevalence and Early Identification of Autism Spectrum Disorder Among Children Aged 8 Years — ADDM Network, 16 sites, United States, 2022*. MMWR, 74(SS-2). — 1 in 31 (3.2%) among 8-year-olds. Community report:

<https://www.cdc.gov/autism/media/pdfs/2025/04/ADDM-Community-Report-SY2022.pdf> · MMWR:

<https://www.cdc.gov/mmwr/volumes/74/ss/ss7402a1.htm>

[5] Lu, Y. et al. (2020). *Prevalence of autism spectrum disorder in Asia: a systematic review and meta-analysis*. Psychiatry Research, 284, 112679. doi:10.1016/j.psychres.2019.112679 (PMID 31735373) — East Asia 0.51%, South Asia 0.31%, West Asia 0.35%.

<https://pubmed.ncbi.nlm.nih.gov/31735373>

[6] Abdallah, B.M. et al. (2025). *Estimates of the prevalence of autism spectrum disorder in the Middle East and North Africa region: a systematic review and meta-analysis*. BMC Public Health, 25:2519. doi:10.1186/s12889-025-23651-x (PMID 40691829) — pooled 0.14% (95% CI 0.02–0.36%); country range 0.01% (Oman) to 2.51% (Saudi Arabia).

<https://pubmed.ncbi.nlm.nih.gov/40691829>

[7] Jiang, X. et al. (2024). *Prevalence of autism spectrum disorder in mainland China over the past 6 years: a systematic review and meta-analysis*. BMC Psychiatry (PMID 38811881) — ~0.7% since 2017 (~1 in 143). Corroborated by a 2024 policy scoping review (SAGE).

<https://pubmed.ncbi.nlm.nih.gov/38811881>

- [8] Song, P. et al. (2021). *The prevalence of adult attention-deficit hyperactivity disorder: a global systematic review and meta-analysis*. General Psychiatry / PMC7916320 — persistent adult ADHD 2.58%; symptomatic adult ADHD 6.76%.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7916320>
- [9] Murray, D., Lesser, M., & Lawson, W. (2005). *Attention, monotropism and the diagnostic criteria for autism*. Autism, 9(2), 139–156.
doi:10.1177/1362361305051398 — origin of the monotropism theory.
<https://doi.org/10.1177/1362361305051398>
- [10] Garau, V., Woods, R., Chown, N., Hallett, S., Murray, F., Wood, R., Murray, A. & Fletcher-Watson, S. (2023). *The Monotropism Questionnaire* (47-item self-report measure; validation preprint). Open Science Framework. — n = 1,110 (756 autistic, 354 non-autistic); autistic mean 4.15 (SD 0.347) vs non-autistic 3.19 (SD 0.578) on a 1–5 scale; ADHD independently predicted higher scores; 8-factor structure. Questionnaire file: <https://osf.io/wpx5g> · validation preprint: <https://osf.io/ft73y> · licence: CC BY-NC-SA 4.0. Local archive: [sources/17-monotropism-questionnaire-osf.pdf](#)